

Vida Adeli

Vector Institute for Artificial Intelligence

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EDUCATION

University Of Toronto (UofT), Toronto, Canada

- Ph.D. in Department of Computer Science (CS) Sep 2021 – Jul 2026
- Thesis: Learning 3D Human Motion under Data Scarcity: Applications to clinical gait analysis and co-speech gesture generation
- Supervisory Committee: Babak Taati, Andrea Iaboni, David Fleet, Animesh Garg

Ferdowsi University of Mashhad (FUM), Mashhad, Iran

- M.Sc. in Artificial Intelligence, Computer Science and Engineering Sep 2015 – Feb 2018
- Thesis: Multi-Stream Human Action Recognition Using Spatiotemporal Saliency Maps
- Supervisors: Prof. Ehsan Fazl-Ersi, Prof. Ahad Harati
- GPA: 19.21 / 20 (With First Rank Honor).

Ferdowsi University of Mashhad (FUM), Mashhad, Iran

- Bachelor of Science in Computer Engineering, Software Sep 2010 – Sep 2014
- Thesis: Object Recognition Using RGB-D data.
- Supervisor: Prof. Ahad Harati
- GPA: 17.10 / 20 (With Second Rank Honor).

Azadegan High School, Mashhad, Iran

- Diploma of Mathematics and Physics Sep 2006 – Jun 2010
- GPA: 19.14 / 20 (With First Rank Honor).

RESEARCH

INTERESTS

Computer Vision, Machine Learning, Image and Video Processing, Deep Learning, Generative Models, 3D Vision, Pose Estimation and Forecasting, Human Activity Recognition, Visual Detection and Recognition, Ambient Intelligence, Healthcare AI Applications.

PUBLICATIONS

Journals

- 2026 V. Adeli, B. Taati, A. Iaboni, “Development and External Validation of a Computer Vision Pipeline for Video-Based Longitudinal Fall Risk Prediction Among Nursing Home Residents with Dementia,” *npj Digital Medicine Preprints (Under review)*.
- 2026 Y. Sharma, V. Adeli, B. Taati, A. Mansfield, A. Mihailidis, A. Iaboni, K. Patterson, “Impact of artificial intelligence on physiotherapist fall predictions in long-term care residents: Observational study,” *JMIR Preprints (Under review)*.
- 2026 B. Filtjens, V. Adeli, S. Mehraban, A. Sabo, B. Taati, A. Iaboni, “Assessment of Change in Drug-Induced Parkinsonism in Daily-Life Gait Using Vision-Based Machine Learning,” *PLOS Digital Health*.
- 2025 Y. Zarghami, M. Muzammil, V. Adeli, H. Reimer, T. Hadjistavropoulos, B. Taati, “PainControl: Identity-Preserving Pain Expression Transfer with Generative Diffusion Models,” *BioMedical Engineering OnLine* (2026).
- 2023 V. Adeli, N. Korhani, B. Taati, A. Iaboni, A. Sabo, S. Mehdizadeh, A. Flint, A. Mansfield, “Vision-based Ambient Monitoring of Gait for Dynamic and Short-Term Falls Risk Assessment in People With Dementia,” *IEEE Journal Of Biomedical And Health Informatics (JBHI)*.
- 2020 V. Adeli, E. Adeli, I., Reid, J. C. Niebles and H. Rezaatofighi, “Socially and Contextually Aware Human Motion and Pose Forecasting,” *IEEE Robotics and Automation Letters (RA-L)*, 5(4), pp.6033-6040.
- 2019 V. Adeli, E. Fazl-Ersi, and A. Harati, “A component-based video content representation for action recognition,” *Image and Vision Computing*, 90, p.103805.

Conferences

- 2027 V. Adeli*, S. Mehraban*, J. Rommann, H. Sanborn, C. Clifford, B. Taati “Splat2Mesh: From Gaussian Splats to Structured and Editable 3D Scenes”, (*To be Submitted*).
- 2027 V. Adeli*, S. Mehraban*, J. Rommann, H. Sanborn, C. Clifford, B. Taati “PUPPETEER: Object-Grounded Posture-Aware Co-Speech Gesture Generation”, (*International Conference on 3D Vision (3DV, Submitted)*).
- 2026 V. Adeli*, S. Mehraban*, J. Rommann, K. Truskovskiy, H. Sanborn, B. Taati, C. Clifford “PickStyle: Video-to-Video Style Transfer with Context-Style Adapters”, (*European Conference on Computer Vision Workshops (Oral)*).
- 2026 V. Adeli, S. Mehraban, M. Mirmehdi, A. Whone, B. Filtjens, A. Dadashzadeh, A. Fasano, A. Iaboni, and B. Taati “GAITGen: Disentangled Motion-Pathology Impaired Gait Generative Model – Bringing Motion Generation to the Clinical Domain”, *Winter Conference on Applications of Computer Vision (WACV 2026)*.
- 2025 V. Adeli, I. Klabučar, J. Rajabi, B. Filtjens, S. Mehraban, D. Wang, H. Seo, T. Hoang, M. Do, C. Muller, C. Oliveira, D. Coelho, P. Ginis, M. Gilat, A. Nieuwboer, J. Spildooren, L. Mckay, H. Kwon, G. Clifford, C. Esper, S. Factor, I. Genias, A. Dadashzadeh, L. Shum, A. Whone, M. Mirmehdi, A. Iaboni, B. Taati “CARE-PD: A Multi-Site Anonymized Clinical Dataset for Parkinson’s Disease Gait Assessment”, *Neural Information Processing Systems (NeurIPS 2025)*. (Website: care-pd.ca)
- 2025 V. Adeli, I. Klabucar, S. Mehraban, A. Iaboni, and B. Taati “Evaluating AI Models for Analyzing Body Movements for Predicting Gait Severity in Parkinson’s Disease: A Cross-dataset Evaluation.”, *International Conference on Aging, Innovation, and Rehabilitation (ICAIR2025)*
- 2024 V. Adeli, S. Mehraban, I. Campose, Y. Zarghami, A. Sabo, A. Iaboni, and B. Taati “Benchmarking Skeleton-based Motion Encoder Models for Clinical Applications: Estimating Parkinson’s Disease Severity in Walking Sequences”, *IEEE International Conference on Automatic Face and Gesture Recognition (FG2024)*.
- 2024 S. Mehraban, V. Adeli, and B. Taati “MotionAGFormer: Enhancing 3D Human Pose Estimation with a Transformer-GCNFormer Network.”, *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV2024)*.
- 2023 C. Malin-Mayor, V. Adeli, A. Sabo, S. Noritsyn, C. Gorodetsky, C. Fasano, A. Iaboni and B. Taati “Pose2Gait: Extracting Gait Features from Monocular Video of Individuals with Dementia.”, *Ambient Intelligence For Healthcare (AmI4HC) International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI2023)*.
- 2021 V. Adeli, M. Ehsanpour, J. C. Niebles, I. Reid, S. Savarese, E. Adeli and H. Rezaatofghi, “TRiPOD: Human Trajectory and Pose Dynamics Forecasting in the Wild,” *IEEE International Conference on Computer Vision (ICCV2021)*.
- 2020 V. Adeli, E. Adeli, I., Reid, J. C. Niebles and H. Rezaatofghi, “Socially and Contextually Aware Human Motion and Pose Forecasting,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS2020)*.
- 2018 V. Adeli, E. Fazl-Ersi, and A. Harati, “Enhancing Human Action Recognition through Temporal Saliency,” *International Conference on Pattern Recognition and Artificial Intelligence (ICPRAI 2018)*, Canada.
- 2018 V. Adeli, E. Fazl-Ersi, and A. Harati, “A New Representation For Human Activity Recognition in Videos Using a Saliency Map,” *4th International Conference on New Studies in Computer and IT*, Mashhad, Iran.

Benchmark and Workshops

- 2026 Workshop on Human Motion Challenges in Real-World and Clinical Settings, [MoCha](#) (ECCV 2026)
+ Benchmark and Challenge on Parkinsonian Gait ([Competition](#))
- 2025 [CARE-PD](#) Benchmark (NeurIPS 2025)
- 2021 SoMoF: SOcial MOtion Forecasting Benchmark, <http://somof.stanford.edu>
- 2021 1st Workshop, Benchmark and Challenge on Human Trajectory and Pose Dynamics Forecasting in the Wild, *ICCV2021*

Abstracts

- 2025 V. Adeli, S. Mehraban, B. Filtjens, A. Whone, M. Mirmehdi, A. Iaboni, B. Taati. "GAITGen: Disentangled Motion-Pathology Impaired Gait Generative Model," *Women in Machine Learning Workshop @ NeurIPS*; 2025.
- 2025 P. Climent-Pérez, V. Adeli, A. Sabo, L. Shum, B. Taati, A. Iaboni. "Computer-vision based ecological assessment of changes in gait in a healthy retirement home cohort," *AGE-WELL Annual Conference*; 2025; Montreal, Canada.
- 2025 V. Adeli, A. Sabo, T. Arora, L. Shum, A. Mansfield, A. Flint, A. Iaboni, B. Taati. "Computer Vision Technology for Falls Risk Prediction In Nursing Home Residents with Dementia," *The Journal of the American Medical Association (TDRA-hosted JAMA Roadshow)*; 2025; Toronto, Canada.
- 2025 V. Adeli, B. Filtjens, S. Mehraban, A. Iaboni, B. Taati. "CARE-PD: A Collaborative Initiative for Harmonizing and Sharing Parkinsonian Gait Datasets to Enable Development of Large-Scale Foundation Models," *Conference Responsible AI in Health Care (CRAIHC)*; 2025; Rotterdam, Netherlands.
- 2025 A. Iaboni, V. Adeli, S. Mehraban, A. Sabo, P. K. Mishra, S. Khan, B. Taati. "Computer vision for monitoring changes in health and behaviour in people with dementia," *Alzheimer's Association International Conference (AAIC) Technology and Dementia Pre-conference*; July 2025; Toronto, Canada.
- 2024 I. Klabucar, V. Adeli, S. Mehraban, B. Taati, and A. Iaboni "Evaluating AI Models for Analyzing body Movements for Predicting Gait Severity in Parkinson's Disease Across Clinical Datasets," *Annual Canadian Movement Disorders Meeting (MDM2024)*.

Posters

- 2023 V. Adeli, S. Mehraban, A. Iaboni and B. Taati. "Evaluating Vision-Based Human Motion Encoders for Gait Analysis and Parkinsonism Severity Prediction in Clinical Settings," *AGE-WELL Annual Conference*; 2023; Toronto, Canada.
- 2023 S. Mehraban, V. Adeli, and B. Taati. "MotionAGFormer: A Versatile Two-Stream Model for Efficient and Accurate 3D Human Pose Estimation in Ambient Monitoring Systems," *AGE-WELL Annual Conference*; 2023; Toronto, Canada.
- 2023 C. Malin-Mayor, V. Adeli, A. Sabo, S. Noritsyn C. Gorodetsky, A. Fasano, A. Iaboni and B. Taati. "Pose2Gait: Extracting Gait Features from Monocular Video of Individuals with Dementia," *AGE-WELL Annual Conference*; 2023; Toronto, Canada.
- 2023 Y. Sharma, V. Adeli, B. Taati, K. Patterson and A. Iaboni. "Evaluating the role of AI-falls risk predictive algorithms in a physiotherapist's falls risk assessment," *AGE-WELL Annual Conference*; 2023; Toronto, Canada.
- 2023 V. Adeli, C. Malin-Mayor, N. Korhani, B. Taati, A. Iaboni, A. Sabo, S. Mehdizadeh, A. Flint, A. Mansfield. "Monitoring Gait with Vision-Based Technology for Short-Term Falls Risk Assessment in Older Adults with Dementia," *The International Conference on Aging, Innovation and Rehabilitation (ICAIR)*; May 8, 2023; Toronto, Canada.
- 2023 C. Malin-Mayor, V. Adeli, A. Sabo, A. Fasano, C. Gorodetsky, A. Iaboni, B. Taati. "Automatically Extracting Gait Features from Video of Older Adults," *The International Conference on Aging, Innovation and Rehabilitation (ICAIR)*; May 8, 2023; Toronto, Canada.
- 2022 C. Malin-Mayor, O. Shalash, A. Sabo, A. Fasano, C. Gorodetsky, V. Adeli, A. Iaboni, B. Taati. "Validating State-of-the-Art 3D Human Pose Tracking for Gait Analysis of Individuals with Parkinson's Disease," *AGE-WELL Annual Conference*; October 18-20, 2022; Regina, Saskatchewan.
- 2022 C. Malin-Mayor, V. Adeli, A. Sabo, A. Fasano, C. Gorodetsky, A. Iaboni, B. Taati. "Automatically Extracting Gait Features from Video of Older Adults," *AGE-WELL Annual Conference*; October 18-20, 2022; Regina, Saskatchewan.

Talks

- 2021 "Towards Human Trajectory and Pose Dynamics Forecasting in the Wild", December 2021, Stanford Vision and Learning Lab (SVL), *Stanford University*.

Magazines

- 2018 V. Adeli, "A friendly introduction to Convolutional Neural Networks (CNNs)," *FUM-ESCAPE Issue*, Scientific Issue of Computer Science Department of Ferdowsi University of Mashhad, Iran.

	• APS360, Applied Fundamentals of Deep Learning (Winter, Summer and Fall 2023, Winter and Summer 2024)	
	Teaching Assistant - University of Toronto, Computer Science Department • CSC420, Introduction to Image Understanding (Fall 2022, 2023, 2024 & 2025) • CSC2503, Foundations of Computational Vision (Winter 2023) • CSC320, Introduction to Visual Computing (Winter 2024)	2022 - Present
	Workshops and Academic Tutorial Classes ▪ Instructor - FUM • Image Processing and MATLAB training classes. • Python Programming Language for Artificial Intelligence Applications ▪ Instructor - FUM, International Campus • Image Processing and MATLAB training classes. ★The course was held for foreign (non-Persian speaking) students in English.	2016 - 2018 2017
	Teaching Assistant of Graduate Courses - FUM - Computer Engineering Department (2016 - 2019) - International Campus (2018) ▪ Computer Vision - Advanced Artificial intelligence - Probabilistic Graphical Models - Deep Learning for Computer Vision - Introduction to Machine Learning - Advanced Image Processing	
	Teaching Assistant of Undergraduate Courses - FUM - Computer Engineering Department (2013 - 2015, 2018) ▪ Computational Intelligence - Fundamentals Of Machine Vision - Theory of Formal Languages and Automata - Compiler Design	
	English Instructor to the students of English as a Foreign Language (EFL), Ferdowsi Language Institute, Mashhad, Iran	Jun 2008 – Oct 2010
R & D EXPERIENCES	PhD Research Scientist Intern , Pickford Technologies, Inc , USA (remote).	May 2025 – Present
	Faculty Affiliate Researcher , Vector Institute , Canada	May 2023 – Present
	Research Assistant , Toronto Rehabilitation Institute (UHN KITE), Canada ▪ Reference: Prof. Babak Taati	Sep 2021 – Present
	Research Assistant , Vision & Learning for Autonomous AI (VL4AI) Lab, Monash University ▪ Reference: Prof. Hamid Rezaatofighi	Aug 2019 – Sep 2021
	Research Associate , Machine Vision Lab, Ferdowsi University of Mashhad, Iran ▪ AI Scientist, OcularAI Inc., Toronto, Canada ▪ Reference: Prof. Ehsan Fazl-Ersi	Feb 2018 – Aug 2021
	Research Assistant , Machine Vision Lab, Ferdowsi University of Mashhad, Iran ▪ Reference: Prof. Ehsan Fazl-Ersi	Sep 2015 – Feb 2018
	Senior Software Developer and Project Lead , Hamta Rayaneh Co., Mashhad, Iran ▪ Reference: Prof. Saeed Abrishami	Mar 2015 – Mar 2016 and Summer 2013
EXECUTIVE EXPERIENCES	VP of Events , KITE Trainee Executive Committee, UHN-KITE, Toronto, Canada Organizing Committee , 1st Workshop, Benchmark and Challenge on Human Trajectory and Pose Dynamics Forecasting in the Wild , (ICCV2021) Member of Executive and Organizing Committee , International Conference on Computer and Knowledge Engineering (ICCKE 2016, 2017, 2018) Executive Committee Chair of Programming Contests Center (ACM) , FUM	Sep 2022 – Present 2014
	Program Committee , 3rd Workshop on Visual Perception for Navigation in Human Environments : The JackRabbit Human Body Pose Dataset and Benchmark (ECCV2022) Member of Reviewing Committee , IEEE Conference on Computer Vision and Pattern Recognition (CVPR)	
COMMITTEE AND MEMBERSHIP		

Member of Reviewing Committee, IEEE International Conference on Computer Vision (ICCV)

Member of Reviewing Committee, IEEE European Conference on Computer Vision (ECCV)

Member of Reviewing Committee, IEEE International Conference on Intelligent Robots and Systems (IROS)

**HONORS &
AWARDS**

- **Best Presentation Award, AI and Dementia International Workshop, AGE-WELL Toronto** 2025
- **PPEF - Beatrix "Trixie" Worsley Graduate Scholarship in Computer Science** 2025
- **[Outstanding Reviewer](#), IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** 2022
- **Mount Sinai Hospital Graduate Scholarships in Science and Technology** 2022
- **Best Paper Award, 4th International Conference on New Studies in Computer and IT.** 2018
- **Ranked 1st, Among Students of M.Sc. graduate program of Artificial Intelligence, FUM** 2018
(National and International Campus)
- **Ranked 1st, Among all M.Sc. Students of Department of Computer Engineering, FUM** 2018
- **Ranked 2nd, Among all B.Sc. Students of Department of Computer Engineering, FUM** 2015